

Vedanth Bharadwaj

Backend Software Engineer

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SUMMARY

Backend Software Engineer with 1+ year of production experience building cloud-native applications across SaaS and healthcare domains. Expertise in Python, SQL, AWS, Java, Spring Boot, REST APIs, distributed systems, and scalable backend architecture, delivering secure, high-performance software.

EDUCATION

Arizona State University - Master of Science, Data Science, Analytics & Engineering

Dec 2025

B.N.M. Institute of Technology (VTU) - Bachelor of Engineering, Information Science & Engineering

Jul 2022

PROFESSIONAL EXPERIENCE

AWS Cloud Developer, MindSpark | Mesa, AZ

Feb 2026 - Present

- Owned backend implementation for the Kairos database gateway, developing Python-based AWS Lambda services supporting 21 relational tables across 500+ enterprise workflows.
- Collaborated with architects and product stakeholders to design scalable REST APIs, Aurora PostgreSQL schemas, and validation logic, accelerating feature delivery by 35%.
- Leveraged Claude Code and OpenAI for AI-assisted coding, SQL optimization, debugging, code reviews, and automated testing, improving sprint productivity by 38%.
- Engineered resilient transaction workflows with rollback, retry backoff, and UUID dependency management, processing 30K+ daily transactions while maintaining 100% relational data integrity.
- Implemented unit, integration, and API validation testing using Pytest, Postman, and automated test suites, increasing backend test coverage to 92% while reducing regression defects by 34%.
- Automated deployments using Git, Lambda versioning, and staged release strategies with production validation, reducing deployment failures by 45% and accelerating release cycles 30%.
- Diagnosed Aurora Serverless bottlenecks using CloudWatch Database Insights, optimizing caching and database initialization, reducing infrastructure costs 75% and cold-start latency below 6 seconds.

Software Engineer, Cognizant | India

Nov 2022 - Jul 2023

- Developed Java Spring Boot microservices and 12+ RESTful APIs backed by SQL databases, reducing average response latency by 35% through query optimization and efficient service design.
- Designed normalized relational schemas and integrated backend services with Angular applications, improving transaction reliability to 99.8% while supporting 5+ business workflows across enterprise applications.
- Collaborated within Agile teams using Git and AWS development tools to deliver 20+ production-ready user stories, resolving 45+ defects through comprehensive integration testing and code reviews.

Software Engineer Intern, Cognizant | India

Mar 2022 - Sep 2022

- Completed an 6-month enterprise engineering program covering Java, Spring Boot, Angular, AWS, SQL, REST APIs, and microservices, maintaining a 95% average across 20+ technical assessments.
- Delivered 3 full-stack healthcare capstone applications by gathering requirements, designing relational schemas, defining REST API contracts, implementing microservices, and performing end-to-end integration testing.

PROJECTS

RAG Document QA Pipeline (Personal)

May 2026

- Designed a production-style RAG platform using FastAPI, LangChain, and hybrid vector databases, reducing hallucinated responses by 40%, improving retrieval precision by 35%, and supporting semantic search across 5,000+ indexed documents.

Urban Mobility Data Pipeline

Dec 2025

- Architected a serverless AWS data platform processing 41M+ transportation records, implementing automated validation, optimized ETL workflows, and scalable storage strategies that improved data quality to 96% while reducing processing time by 65%.

Energy Consumption Forecasting

May 2024

- Developed a scalable energy analytics platform exposing forecasting services through REST APIs, integrating automated preprocessing, feature versioning, and optimized inference pipelines supporting 1M+ records while reducing prediction latency from 310 ms to 180 ms.

CERTIFICATIONS

AI Fluency - Framework & Foundations, Anthropic

Google Data Analytics Professional Certificate (SQL, Tableau, R, Data Cleaning), Google

Data Analysis and Visualization with Power BI, Microsoft

SKILLS

Programming Language : Python, SQL, Java, JavaScript, TypeScript

Backend : FastAPI, Spring Boot, REST APIs, Microservices, Distributed Systems, Backend Architecture

Cloud & DevOps : AWS (Lambda, API Gateway, Aurora PostgreSQL, S3, DynamoDB, CloudWatch), Git, Docker, Maven

Databases : PostgreSQL, MySQL, Aurora PostgreSQL, DynamoDB, RDS Data API

AI-Assisted Development : Claude Code, OpenAI

AI & Retrieval : LangChain, FAISS, ChromaDB, RAG, Sentence Transformers

Software Engineering : Object-Oriented Design, Data Structures & Algorithms, System Design, CI/CD, Agile, SDLC